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Modelling and numerical simulation of *n*-heptane pyrolysis coking characteristics in a millimetre-sized tube reactor

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ABSTRACT

Based on the optimization of the recently proposed gas phase mechanism for *n*-heptane pyrolysis, mechanisms for polycyclic aromatic hydrocarbon (PAH) formation and a detailed wall surface reaction, a basic gaseous pyrolysis mechanism with 1127 steps and a simplified surface reaction mechanism with 34 steps were developed, respectively. After performing the detailed estimation and modelling procedures for the surface site density (*SDEN*) related to the particle diameter, particle density, C/H ratio and other parameters determined by the experimental tests and observations, numerical studies were integrated with the developed coke model to simulate the pyrolysis process and the coking properties in a millimetre-sized tube reactor on the Chemkin Pro software platform using a two-dimensional cylindrical shear flow (CSF) module. The prediction values of the pyrolysis product concentration and coking rate at different temperatures were compared with the experimental data, and agreement was obtained. Under pyrolysis conditions of an *n*-heptane inlet flow rate of 0.5 ml/min, an operating pressure of 1.0 MPa and a furnace temperature of 973–1073 K, the research results showed that the hydrogen abstraction reaction, as the first and key step of coke formation, is easily affected by gaseous diffusion. As the temperature and particle diameter increase, the inhibition of diffusion becomes stronger. Among all the considered radicals, the abstraction ability of the H radical is the strongest, and it is easily restricted by diffusion. The presence of abundant radicals such as CH_3, C_2H_3, and C_3H_5 at low temperatures can compensate for this limitation. For the unsaturated species addition reactions, the calculation results showed that the addition of olefins, alkynes, diolefins and aromatic hydrocarbons occurs in sequence with an increase in residence time. With increasing temperature, the addition contributions of olefins such as C_2H_4 and C_3H_6 gradually weaken, while those of small molecule aromatic hydrocarbons such as benzene (A1) and styrene ($A1C_2H_3$) remarkably increase. Alkynes and diolefins, such as C_2H_2 and C_3H_4, have low concentrations under the experimental conditions, and their addition contributions are very low and can be ignored. Due to the addition of unsaturated species, the concentration of H_2 in the gas phase is significantly improved. The modeling work can predict the coke formation characteristics and quantitatively evaluate the change in the coke formation rate, which is helpful for explaining the intrinsic thermal pyrolysis coking mechanism in a round tube, such as the regenerative cooling system of supersonic engines, among others.

1. Introduction

Thermal management of advanced hypersonic flight vehicles has been a key issue for several decades. To ensure operation reliability and durability, a regenerative cooling system is a feasible solution that can be applied in supersonic combustion ramjet or scramjet engines. In the regenerative cooling system, hydrocarbon fuels are decomposed to small molecule gases due to the large heat flux from the combustion chamber wall, and then these gases enter the combustion chamber. The extra chemical heat sink released from the pyrolysis process absorbs the heat flux quickly and protects the wall material. The pyrolysis product, such as ethylene, can also enhance the ignition and combustion performance in the chamber [1]. However, coking may occur in these millimeter-sized cooling channels under conditions of high temperature and pressure, which are easily reached in modern aircrafts [2]. Until now, determining how to avoid coking has been challenging; therefore,

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studying the intrinsic mechanism of the coke formation process is essential.

n-Heptane (*n*-C_7H_{16}) is regarded as a primary reference fuel, and it is a representative normal alkane that has been used as a surrogate gasoline fuel for research on combustion processes in internal combustion engines [3]. In the past several decades, many experiments and reaction mechanism studies on the combustion and pyrolysis of *n*-heptane have been carried out extensively [4–8]. The reaction kinetics mechanism developed by Westbrook et al. [9] and Curran et al. [10,11] as well as other mechanisms summarized in [12] have been widely used in hydrocarbon fuel combustion and pyrolysis studies. These studies provide useful information on the mechanism of decomposition reactions and provide reliable theoretical and experimental data for further studies of more complex processes. For predicting the polycyclic aromatic hydrocarbons (PAHs) and soot generation, Frenklach et al. [13] proposed a hydrogen-abstraction–acetylene-addition (HACA) mechanism for PAH growth, which has also been developed and validated in other studies to predict the growth of PAHs into up to five-ring aromatic species [14]. A mechanism for gasoline surrogate fuels (KM1) [15] and its updated version (KM2) [16] was developed from the base mechanism (USC Mechanism [17] for C_1–C_4 fuels) to account for the growth of PAHs up to coronene. Furthermore, some PAH-based soot generation models were also developed and summarized in published literatures [18–20]. In addition to modeling research, many validation experiments have been carried out in combustion devices [21–23], including jet-stirred reactors, shock tubes, counterflow flames and engines.

Compared with the gaseous reaction of PAHs, coking mainly occurs on the wall and is not the same as soot generation in a micro channel. Coke is formed by complex chemical and physical processes of endothermic hydrocarbon fuel, wherein the chemical mechanisms involved include the thermal oxidation mechanism and the pyrolysis coking mechanism [24]. Thermal oxidative deposition occurs at low temperatures (below 753 K) and is ascribed to dissolved oxygen in the fuel, while at relatively high temperatures (above 753 K), pyrolysis deposition becomes dominant [25]. A few studies have modeled pyrolysis coking during the thermal cracking of hydrocarbon fuels. Sheu et al. [26] performed pyrolysis deposition experiments on hydrocarbon fuel and developed a zero-order, surface reaction model in which the wall temperature was the only factor determining the coking rate. Gascoin et al. [27] characterized the coke activity from the thermal cracking of *n*-dodecane (*n*-$C_{12}H_{26}$) and correlated the coke formation rate with the residence time and concentration of methane. Liu et al. [28,29] selected unsaturated hydrocarbons and aromatics as coke precursors and developed a coking kinetics model to calculate the rate of coke formation considering the effects of surface coverage, local precursor concentrations, and wall temperature. Xu et al. [30] simulated the fluid flows and heat transfer of aviation kerosene with consideration of both fuel pyrolysis and surface coking at supercritical pressures. However, in that coking model, the precursor concentration was obtained experimentally, so the model is empirical, and the theoretical relationship between gaseous pyrolysis and the coking process has not yet been determined. Additionally, coke formation is also attributed to the soot formation reactions taking place in the fluid phase and subsequent deposition on the surface [31], which should also be taken into consideration at high temperatures or longer reaction times.

Based on the above studies, it can be summarized that pyrolysis coking sources are mainly divided into two types, which are highlighted briefly in Fig. 1. The total coke formation rate can be regarded as the sum of contributions of the surface reactions and particle deposition. However, under the conditions of temperatures lower than 1123 K and fuel residence times shorter than 5 s in the experiments and simulations in this work, route *I* can be ignored because the temperature or residence time is not sufficiently high to form large PAHs and soot particles, so the surface reaction mechanism involved in route *II* prevails and is the focus of this study. In the past several decades, a few chemical modeling works have been carried out related to the surface reactions between gas species and metal oxides or carbon-based materials. Frenklach et al. [32] proposed a model with 52 irreversible surface elementary steps to predict the deposition kinetics during a chemical vapor deposition (CVD) process. Wauters et al. [33] used a similar approach to model coke formation during steam cracking. Okkerse et al. [34] developed a kinetic mechanism with 67 elementary steps involving 41 species to explain the diamond surface growth in an oxy-acetylene torch reactor. A detailed heterogeneous kinetic mechanism with 275 elementary steps involving 66 surface sites was developed by Fournet et al. [35,36] to model pyrocarbon deposition obtained by propane pyrolysis on carbon fibers. The author found that pyrocarbon seems to be mainly formed by the deposition of small unsaturated species (such as acetylene and ethylene) and methyl radicals. Although the above works were performed with different reactors or devices, the detailed chemical reaction mechanism can provide thorough information and a deeper understanding of the coke process in a tube reactor.

Taking into account both effectiveness and practicality, we select and validate the reaction mechanisms [11,16,35] for *n*-heptane pyrolysis and surface reactions; then, a simplified coke formation model is developed, and simulations for a millimeter-sized tube reactor are performed on the Chemkin Pro software platform. After comparing the results with the experiments, the coking characteristics and their dependence on the temperature and time duration are studied in this work.

2. Experimental setup and measurements

A schematic diagram of the experimental device is shown in Fig. 2. A stainless steel tube with a length of 100 cm and an internal diameter and thickness of 2.18 × 0.5 mm was used as the reactor. The tube was oxidized and passivated at a high temperature (1473 K) for 2 h before being used to eliminate the catalytic effect. The tube was put into an electric heating furnace with a temperature control accuracy of ± 1.0 K. Vacuum filtration of *n*-heptane with a 0.22 μm organic membrane was used to remove the dissolved oxygen and tiny impurities in the liquid. The high pressure infusion pump was used to precisely control the injection flow rate. Before the sample was placed in the reactor, the reactor was purged with nitrogen for 10 min to exclude the air. To prevent the influence of the *n*-heptane phase change on the pyrolysis process, the fuel was preheated to a constant temperature of 573 K and completely vaporized before it entered the reactor. A back pressure valve was used to adjust the pressure in the entire reaction channel. The outlet products were rapidly cooled below 353 K to reduce the secondary reaction and dissolved gas in the liquid, and then the products were transferred to an ice–water bath for gas–liquid separation. Considering that the difficulty of directly measuring the fluid temperature arises from the serious coking effect, thermocouple wires were welded on the outer wall surface of the tube at 10 cm intervals to monitor the temperature change. The products were collected and analyzed after the temperature was stable for each reaction condition. The gas phase was analyzed by an Agilent 7890A, and the liquid phase was analyzed by TRACE 1300ISQ gas chromatography-mass spectrometry.

The reason for selecting the coking conditions was that the effective diameter did not greatly change as a result of the coking process; therefore, there was no significant effect on the normal flow in the tube, while at the same time, the amount of generated coke was sufficient to reduce the error in weighing. After detailed preliminary experiments, it was determined that the proper

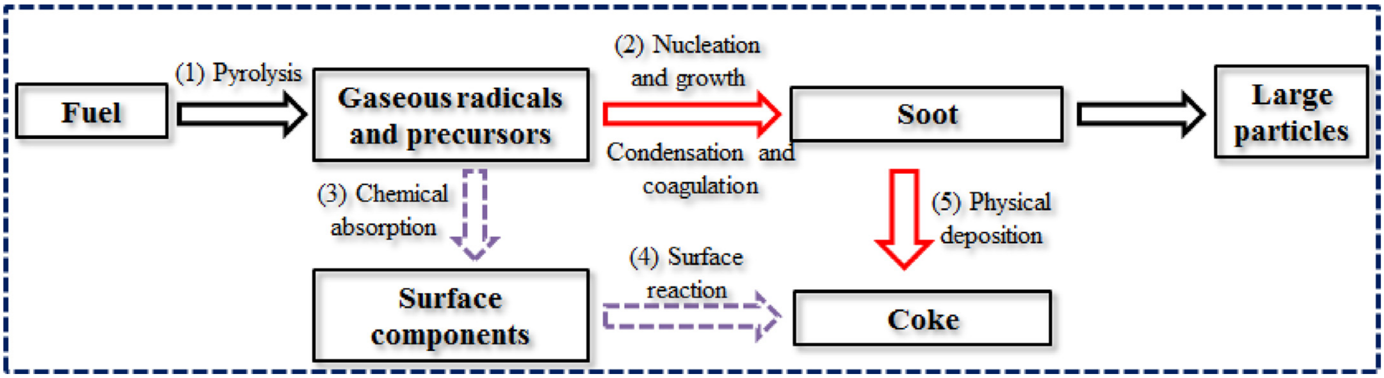


Fig. 1. Schematic diagram of *n*-heptane pyrolysis and coking mode in a tube. *Route I: Through dehydrogenation, condensation and other chemical reactions, small molecules produced from the vaporized fuel pyrolysis react to become aromatic hydrocarbons, polycyclic aromatic hydrocarbons, and soot. Then, they are deposited on the surface in a process called soot deposition, as shown as sub-processes (1), (2) and (5). Route II: The vapour phase precursors, such as olefins, alkynes, or small molecules of PAHs, can be adsorbed on the channel surface active sites and react with the carbon nuclei. This process accelerates the formation of coke and growth of the carbon layer, which is called a surface reaction, as shown as sub-processes (1), (3) and (4).*

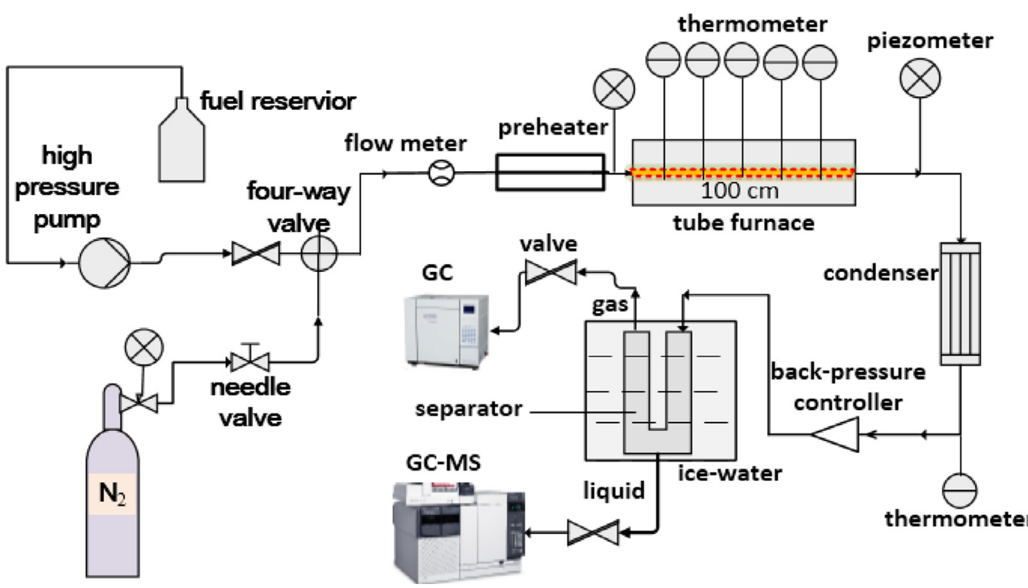


Fig. 2. Schematic diagram of the experimental device.

coking experimental conditions were an *n*-heptane inlet flow rate of 0.5 ml/min, an inlet temperature of 573 K, an operating pressure of 1.0 MPa and a furnace temperature of 973–1073 K. Under the selected conditions, the monitored temperature and pressure remained relatively stable. Each experiment lasted for 80 min, and then the total mass of the tube was weighed by an electronic balance with an accuracy of 0.1 mg. The formation coke mass can be obtained after deducting the initial value of the blank tube. Taking into account the instability of the coking process, each experiment was repeated five times and then averaged. The morphology and structure of the sampled particles were observed by scanning electron microscopy (SEM), and the coke C/H ratios at different temperatures were measured with an elemental analyzer.

3. Kinetic model expressions

3.1. Mechanism of thermal pyrolysis and PAH formation

The gaseous reaction kinetics mechanism with C_1–C_7 species of National University of Ireland Galway (NUIG) [11] and the PAH formation kinetics mechanism KM2 with seven-ring aromatic species [16] were selected in this work. The NUIG mechanism has considered many fall-off effects on the pressure-dependence reactions and has modified the rate constants with the change in pressure. The KM2 mechanism was based on the reaction mode of HACA, and several reactions involving odd carbon number species, such as indenyl (C_9H_7) and cyclopentadienyl (C_5H_5), contributed to the synergistic effects on PAH formation. The KM2 mechanism can predict PAH growth from benzene (A1) to coronene (A7). To simulate the pyrolysis conditions, we extracted and eliminated the species and reactions including oxygen from the entire mechanism. After the combination and assembly procedure, a basis for the gas–solid reaction kinetics mechanism for *n*-heptane pyrolysis was obtained.

It should be noted that the above mechanisms were mainly developed based on high-temperature combustion conditions, and the adaptability to the low-temperature pyrolysis process still needs to be validated. In this paper, adjustments and supplements have been made to the original reaction mechanism based on the following analysis.

I. The first C–C bond dissociation reaction of *n*-heptane has a highly significant effect on the pyrolysis rate and small molecule production [37]. Under the combustion condition at relatively high temperatures, the influences of the attack of oxygenated radicals and the oxidization reaction are far greater than that of the breaking of carbon bonds, meaning that the

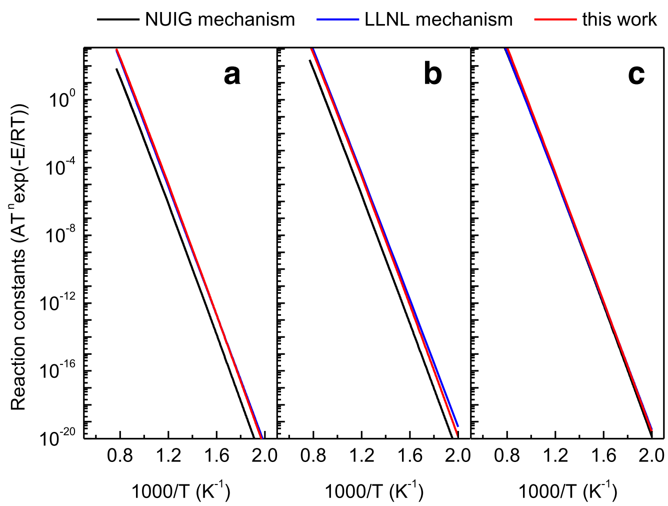


Fig. 3. Rate constants of the first C–C bond dissociation reaction of *n*-heptane. (a) $n\text{-}C_7H_{16} = CH_3 + C_6H_{13}$; (b) $n\text{-}C_7H_{16} = C_2H_5 + C_5H_{11}$; (c) $n\text{-}C_7H_{16} = n\text{-}C_4H_9 + n\text{-}C_3H_7$.

latter makes little contribution to the *n*-heptane consumption. In contrast, under pyrolysis conditions, the oxidization path disappears, and the sensitivity of C–C bond dissociation increases to a great extent and controls small radical production and consequent reactions. Li et al. [38] obtained the path and reaction constant of *n*-heptane decomposition by quantum calculations in parallel with reactive force field MD simulations (ReaxFF) and compared them with the commonly used reaction kinetics mechanisms. To further confirm this hypothesis, detailed heptane pyrolysis experiments were performed in our previous work [39]. As a result, we correspondingly adjusted the pre-exponential factor of the first step of the *n*-heptane pyrolysis reaction in the NUIG mechanism, and the modified reaction constants are very close to the Lawrence Livermore National Laboratory (LLNL) mechanism [40], as shown in Fig. 3.

II. Benzene (A1) and toluene ($A1CH_3$), as intermediate products of pyrolysis, are generally considered important precursors for PAHs and soot generation. According to the sensitivity and formation rate analysis of benzene, acetylene (C_2H_2) and the propargyl radical (C_3H_3) are important intermediates under combustion conditions. Benzene is mainly formed through the $C_3H_3 + C_3H_3$ reaction pathway. However, the formation of C_2H_2 is inhibited under pyrolysis conditions with lower temperatures. The reaction of $C_2H_3 + C_4H_6 = A1 + H_2 + H$ significantly affects the formation of the initial benzene ring, which is different from the main formation path of benzene under combustion conditions, as shown in Fig. 4. The above reaction has been studied in detail by Orme et al. [41] and is included in the KM2 mechanism. Additionally, Wang et al. [42] found that the Diels–Alder reactions of ethylene (C_2H_4) and 1,3-butadiene (C_4H_6), C_2H_4 and methyl-1,3-butadiene (C_5H_8) are also very significant under pyrolysis conditions. The continuous dehydrogenation of the resulting cyclohexene (C_6H_{10}) and methyl cyclohexene (C_7H_{12}) is another important route to benzene and toluene production, respectively. Taking into account the complexity of the benzene generation path under low-temperature pyrolysis conditions, this work also added some important reactions related to cyclohexene and methyl cyclohexene selected from previous works [41,42], thereby enriching the formation path of benzene and toluene and improving the simulation accuracy of their production rate under low temperature and pyrolysis condition.

Through the above adjustments, we obtained a basic mechanism with 230 elements and 1127 chemical reactions for gaseous pyrolysis (*detailed information is shown in Supplementary Materials I*). Using this mechanism, the reaction field parameters and the gaseous species concentration can be obtained for the subsequent coking calculation.

3.2. Mechanism of the wall surface reaction

The process of radical absorption and surface reactions (3) and (4), as shown in Fig. 1, is related to the type of adsorbent material and the wall properties. When the surface is smooth and the active sites formed by the metal elements are occupied, there are many small scattered spherical nuclei formed through a so-called surface catalytic coking mechanism, which has been studied and reported in previous works [24–28]. In this study, we only focused on the pyrolysis coking properties after the catalytic coking was almost complete. At that time, the continuing coking process will occur on the surface composed of the active hydrocarbon-containing mixtures rather than the smooth wall itself. Without considering the diffusion limit, and under the chemical kinetics-controlled regime, the coke particle growth process is determined by several kinds of reactions, such as H abstraction, the addition of unsaturated species, termination reaction between two neighboring surface sites, etc., similar to the growth of gaseous aromatic species or soot particles. Therefore, for each surface reaction, an elementary gaseous process chemically close to the surface process can be chosen as the "prototype" reaction. Many previous works [18,35,36] have suggested that activation energies and pre-exponential factors are equivalent for prototype and surface reactions. Based on the above treatment method, the surface reaction steps and their reaction constants were adopted and are described as follows:

3.2.1. H abstraction reactions

For the pyrolysis coking process, the first step of unsaturated species deposition is the formation of an active carbon radical site by the hydrogen abstraction reaction:

$$R^* + H(s) \Rightarrow open(s) + RH$$
$$RH + open(s) \Rightarrow R^* + H(s) \quad (1)$$

where the symbol (s) represents the surface species, H(s) represents the H site fraction on the particle surface, open(s) represents the unoccupied fraction of active sites, and R^* represents different radicals in the gaseous phase around the particle. For PAH formation, the KM2 mechanism gives the H abstraction reaction from PAH species via reactions with H, CH_3, C_2H and C_3H_3 radicals [43-45]. In addition to these radicals, Fournet et al. [35] also considered the effects of C_2H_3, C_2H_5, iC_4H_3, aC_3H_5, benzyl and indenyl on the H abstraction in the CVD process. In this work, based on the "prototype" hypothesis, we selected the rate constants of abstractions by the radicals H, CH_3 and C_3H_3 from the KM2 mechanism [16] and assumed the rate on the particle surface to be the same as the rates for pyrene (A4), considering that the chemical structures of larger PAHs are closer to the coke surface than those of small PAHs, such as benzene or naphthalene. For other main radicals in the gas phase, their H abstraction reaction rates were selected from Fournet's work [35].

3.2.2. Unsaturated species addition reactions

C_2H_2 was regarded as the dominant addition species in the original HACA mechanism [13]. Many researchers have proven the feasibility of this mechanism for different flames, especially in regimes with high temperatures, and this mechanism has been used comprehensively in the modeling of soot formation [18,19]. Under the pyrolysis conditions of 873–1073 K, we found that

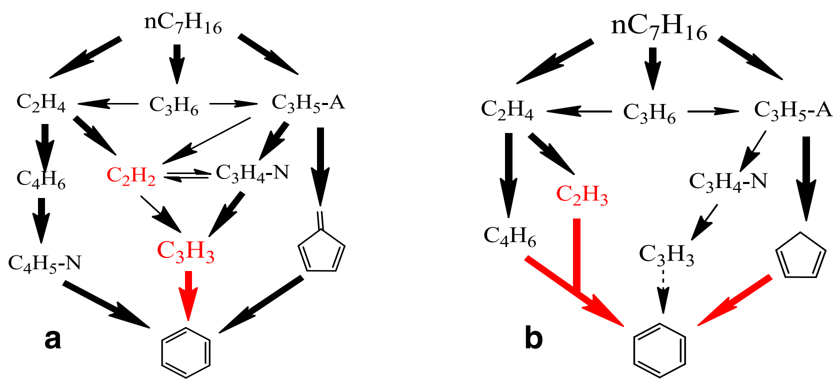


Fig. 4. Main reaction paths for benzene under (a) combustion and (b) pyrolysis conditions.

alkynes, such as C_2H_2, have very low concentrations, while some olefins, such as the C_2H_4 fraction, can reach contents of 20–30%. Correspondingly, the contents of aromatic compounds with alkenyl groups such as styrene ($A1C_2H_3$) are much higher than the contents of aromatic compounds with alkynyl groups such as phenylacetylene ($A1C_2H$), showing that the vinyl radical (C_2H_3) has a very important contribution to the formation of benzene and phenyl compounds. Under pyrolysis conditions, some alkenes and alkenyls may act effectively on the formation and growth of aromatic ring structures due to the Diels–Alder addition and macromolecular polymerization, thereby increasing the mass yield rate of solid particles. The previous experimental results by Raman and infrared spectroscopies [46] showed that the soot structure is covered by a large amount of aliphatic substances with alkenyl groups, especially at the early stage of soot growth and under relatively low-temperature conditions, indicating that more surface reaction paths should be supplied.

As a CVD-like process, the conditions of pyrolysis coking on the wall surface mainly occur in the absence of oxygen and under moderate temperature conditions. The deposition of gaseous molecules containing triple bonds (acetylene (C_2H_2), propyne (PC_3H_4) and phenylacetylene ($A1C_2H$)) and double bonds (ethylene (C_2H_4), propene (C_3H_6), propadiene (AC_3H_4), styrene ($A1C_2H_3$), benzene (A1) and naphthalene (A2)) can occur once the active sites, designated as open(s), are created. All these chemical reactions are assumed to occur on the edges of “basic structural units” on both armchair and zig-zag sites [35]. Based on their detailed mechanisms, the different surface components with aliphatic groups, such as CH_3(s), C≡CH(s), CH=CH_2(s), which have been detected experimentally [47–48], can be quantitatively calculated.

However, we focus on the coke formation rate and its change at different temperatures, so Fournet’s mechanism can be simplified to suit the requirements of this work. Based on the rate of production (ROP) analysis and our experimental results, the following simplifications were made. First, compared with the armchair site direct deposition by single unsaturated species, we ignored the zig-zag site deposition since the addition of at least three carbon atoms is required to complete the formation of a six-atom ring with the combination of $C_2H_2+CH_3$, $C_2H_4+CH_3$, $CH_3+CH_3+CH_3$, etc., so the reaction rate of zig-zag site deposition is relatively unimportant, especially when the equilibrium concentrations of radicals and C_2H_2 are very low based on the pathway analysis [36]. Second, the rates of the termination reaction between two neighboring surface sites and the dehydrogenation of aliphatic surface sites are relatively low, and these rates control the distribution of C and H atoms in different surface sites. However, coke is a mixture, and the total coking rate should include all active sites with carbon atoms, such as CH_3(s), C≡CH(s), and CH=CH_2(s), and it depends on the first addition step of unsaturated species. That is, whether or not the carbon and hydrogen atoms in the aliphatic surface sites are separated entirely, their total mass can still contribute to coke growth. Based on the above reasons, we used an irreversible reaction to synthetically describe the effect of the unsaturated species addition on the coke formation process:

$$M + open(s) \Rightarrow H(s) + H_2 + xC_B + yH \tag{2}$$

where M represents the added gaseous species, C_B represents the formed coke, and x and y are the reaction coefficients calculated from the mass balance. The overall surface reactions (2) can be described as follows. After abstraction of the radical, as shown in reaction (1), an active site open(s) is created; species M adds on to this site and leaves a new H(s) for continuous addition, and the excessive hydrogen components are released into the gas phase in the form of hydrogen and hydrogen radicals, respectively. The rate constant of reaction (2) for different species is regarded as that of the first addition step in the detailed addition mechanism [35].

Based on the above consideration and analysis, the simplified surface reaction mechanism includes 2 surface sites, and 34 reaction steps were developed, as listed in Table 1 (*detailed information is shown in Supplementary Materials II*). It should be emphasized that this mechanism is used only to predict the total coke formation rate in the current work, and it decreases the calculation time while ensuring the accuracy and increasing the conciseness of the whole analysis.

3.3. Estimation of surface site density under a chemical kinetics-controlled regime

Richter et al. [49] determined the characteristics of larger soot particles, such as molecular weight, C/H ratio and diameter, based on the assumption of spherical structures and a density of 1.8 gcm^{-3}. In a particle growth process, the number density of the H atom on the outer surface can be regarded as the surface site density (*SDEN*), which can be expressed as follows:

$$SDEN_p = \frac{\rho_p d_p}{6(12f+1)} \tag{3}$$

where ρ_p, d_p and f represent the particle density, diameter and C/H mole ratio, respectively. A comparison of the calculation results from formula (3) and the data in [49] is shown in Fig. 5, and good agreement is observed. As a similar process, the *SDEN* of a single growing particle on the surface can also be estimated by formula (3).

However, in the actual coke process, many scattered spherical particles initially form. As the particle diameter continues to increase, surface growth exhibits anisotropy due to the non-uniform distribution of gaseous precursors and the influence of other particles, which induce the presence of new particles at new locations on the surface, leading to “spatial growth”, as shown in Fig. 6(a). At the same time, the increases in the particle number and size

Table 1
Simplified surface reaction mechanism.

H abstraction reactions		A	n	E
$H + H(se) \rightarrow open(se) + H_2$	(R1)	4.9E8	1.884	9.829E3
$open(se) + H_2 \rightarrow H + H(se)$	(R2)	4.9E4	2.467	2.926E3
$H(se) \rightarrow open(se) + H$	(R3)	1.78E16	0.0	1.125E5
$open(se) + H \rightarrow H(se)$	(R4)	2.0E13	0.0	0E0
$CH_3 + H(se) \rightarrow open(se) + CH_4$	(R5)	7.98E-1	3.933	1.177E4
$open(se) + CH_4 \rightarrow CH_3 + H(se)$	(R6)	4.48E-2	4.248	4.277E3
$C_3H_3 + H(se) \rightarrow open(se) + AC_3H_4$	(R7)	1.908E1	3.529	2.445E4
$open(se) + AC_3H_4 \rightarrow H(se) + C_3H_3$	(R8)	1.36E0	3.761	1.089E3
$C_2H + H(se) \rightarrow open(se) + C_2H_2$	(R9)	2.0E13	0.0	0E0
$C_2H_3 + H(se) \rightarrow open(se) + C_2H_4$	(R10)	1.198E-1	4.0	7.7E3
$open(se) + C_2H_4 \rightarrow H(se) + C_2H_3$	(R11)	4.72E11	0.0	9.866E3
$C_2H_5 + H(se) \rightarrow open(se) + C_2H_6$	(R12)	1.0E11	0.0	1.5E4
$open(se) + C_2H_6 \rightarrow H(se) + C_2H_5$	(R13)	5.1E11	0.0	2.608E3
$C_3H_5\text{-}A + H(se) \rightarrow open(se) + C_3H_6$	(R14)	1.05E11	0.0	2.0E4
$open(se) + C_3H_6 \rightarrow H(se) + C_3H_5\text{-}A$	(R15)	4.54E10	0.0	−5.888E3
$i\text{-}C_4H_3 + H(se) \rightarrow open(se) + C_4H_4$	(R16)	1.05E11	0.0	2E4
$open(se) + C_4H_4 \rightarrow H(se) + i\text{-}C_4H_3$	(R17)	8.01E12	0.0	−6.2E2
$C_6H_5CH_2 + H(se) \rightarrow open(se) + C_6H_5CH_3$	(R18)	7.63E13	0.0	3.587E4
$open(se) + C_6H_5CH_3 \rightarrow H(se) + C_6H_5CH_2$	(R19)	7.90E13	0.0	1.2E4
$C_9H_7 + H(se) \rightarrow open(se) + C_9H_8$	(R20)	6.4E13	0.0	4.264E4
$open(se) + C_9H_8 \rightarrow H(se) + C_9H_7$	(R21)	1.0E14	0.0	1.2E4
Species addition reactions		**A**	**n**	**E**
$open(se) + C_2H_2 \rightarrow H(se) + 2C(B) + H$	(R22)	8.0E7	1.56	3.8E3
$open(se) + C_2H_4 \rightarrow H(se) + 2C(B) + H + H_2$	(R23)	1.25E12	0.0	6.2E3
$open(se) + PC_3H_4 \rightarrow H(se) + 2C(B) + CH_3$	(R24)	8.3E22	−2.7	1.747E4
$open(se) + AC_3H_4 \rightarrow H(se) + 2C(B) + CH_3$	(R25)	2.5E12	0.0	6.2E3
$open(se) + C_3H_6 \rightarrow H(se) + 2C(B) + CH_3 + H_2$	(R26)	2.5E12	0.0	6.2E3
$open(se) + A_1 \rightarrow H(se) + 6C(B) + H_2 + 3H$	(R27)	3.26E12	0.0	6.4E3
$open(se) + A_2 \rightarrow H(se) + 10C(B) + H_2 + 5H$	(R28)	3.26E12	0.0	6.4E3
$open(se) + A_1C_2H \rightarrow H(se) + 8C(B) + 5H$	(R29)	8.3E22	−2.7	1.747E4
$open(se) + A_1C_2H_3 \rightarrow H(se) + 8C(B) + 5H + H_2$	(R30)	2.5E12	0.0	6.2E3
$open(se) + C_6H_5 \rightarrow H(se) + 6C(B) + 4H$	(R31)	3.0E12	0.0	0E0
$open(se) + C_2H_3 \rightarrow H(se) + 2C(B) + H_2$	(R32)	5.0E12	0.0	0E0
$open(se) + C_2H \rightarrow H(se) + 2C(B)$	(R33)	2.81E12	0.0	0E0
$open(se) + CH_3 \rightarrow H(se) + C(B) + H_2$	(R34)	3.0E13	0.0	0E0

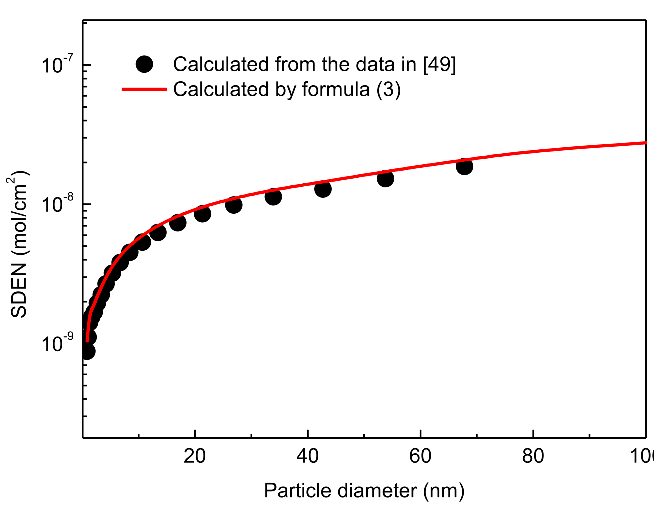


Fig. 5. Change in the surface site density (*SDEN*) with a single particle diameter.

also promote mutual fusion (Fig. 6(b)), and the result is that the coke body gradually covers the wall induced by a wide range of particle fusion and the mode similar to "planar growth" takes place (Fig. 6(c)). Same growth processes have been observed by many researchers [28–29, 50–51]. The growth of the coke on the wall has clearly undergone three stages: spatial growth, planar growth and a transition stage.

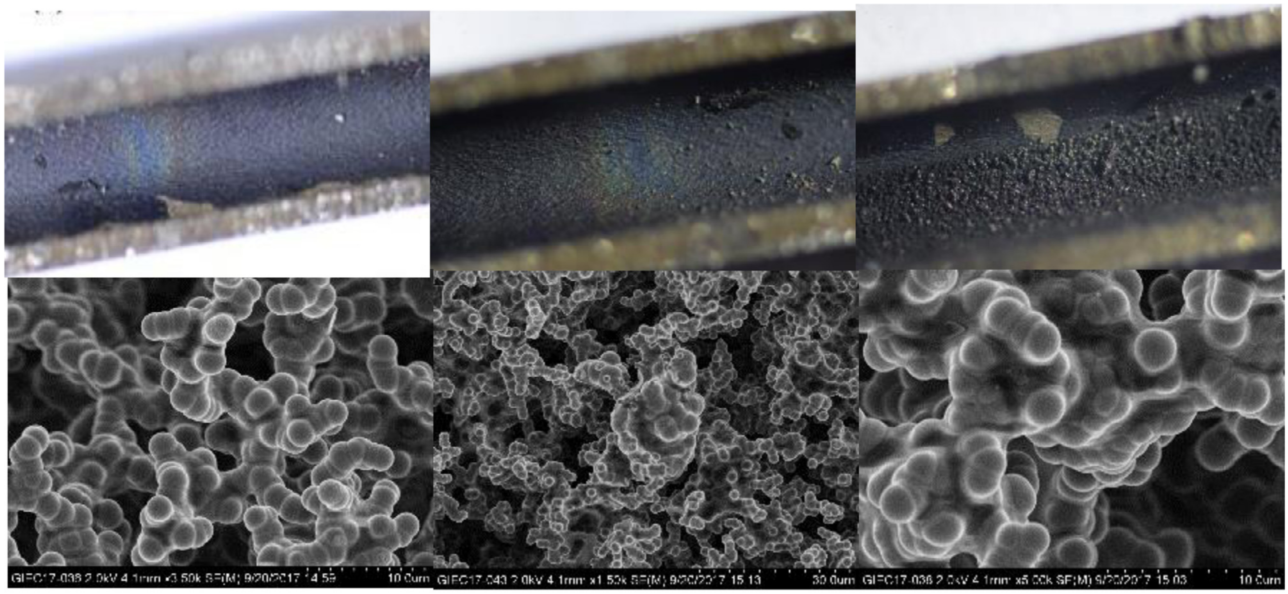
Considering that the active site changes from the spatial growth stage to the planar growth stage, we assume that there is a critical diameter, d_{pcrit}, between the two modes of transformation. Before reaching this critical diameter, the active site density increases with the particle size according to the relationship shown in formula (3). Consequently, the diameter of a single particle increases very slowly, and particle mutual fusion takes place. The active site density gradually decreases to a stable value due to the disappearance of the spatial growth mode. Under complete chemical controlling conditions and as a modeling treatment, the moment when the particles make contact with each other is the moment with the largest active site density. However, it is complicated to fully describe the entire non-steady process, which is required to solve the gas–solid two-phase iterative equation. As a preliminary model study based on experimental phenomena in this work, we use a piecewise linearization method in which the change in the active site density during each process is replaced by the time-average value. Although this processing method is relatively simple, it is proven to be effective according to the experimental results, as introduced in the next Section 4.1. After derivation, the average active site density based on the wall surface area (which is not the same as the site density based on the particle surface area, $SDEN_p$) at different stages is expressed as follows:

$$\overline{SDEN}_{w-s} = \frac{\frac{\pi}{4}\rho_p d_{pcrit}}{6(12 f_{pcrit} + 1)} \quad \text{(at the spatial growth stage)} \tag{4}$$

$$\overline{SDEN}_{w-sp} = \frac{\frac{\pi+1}{2}\rho_p d_{pcrit}}{6(12 f_{pcrit} + 1)} \quad \text{(at the transition stage)} \tag{5}$$

$$SDEN_{w-p} = \frac{\rho_p d_{pcrit}}{6(12 f_{pcrit} + 1)} \quad \text{(at the planar growth stage)} \tag{6}$$

From the beginning of formation to the gradual stabilization of the surface, if assuming the same time interval for every stage, the



(a) Spatial growth (b) Transition stage (c) Planar growth

Fig. 6. Particle morphology at different coking stages.

integrated average site density during the whole process is estimated as:

$$SDEN_{w-ave} = \frac{\overline{SDEN}_{w-s} + \overline{SDEN}_{w-sp} + SDEN_{w-p}}{3} = \frac{\left(\frac{\pi}{4} + \frac{1}{2}\right)\rho_p d_{pcrit}}{6(12 f_{pcrit} + 1)} \quad (7)$$

It should be emphasized that formulas (4)–(7) are derived on the basis that the "final planar growth" assumption is met. For a round tube, this assumption can only be established if the tube diameter d remains approximately constant and much larger than the coke diameter d_p. When the pipe diameter is extremely small or the pipe is almost blocked, the instantaneous value of the critical diameter d_{pcrit} will vary greatly due to the centripetal structure of a round tube. To avoid this variation, the selected experimental conditions are an n-heptane inlet flow rate of 0.5 ml/min, an operating pressure of 1.0 MPa, a temperature from 973 to 1073 K, and a pyrolysis time of 80 min, as mentioned in Section 2. Under these conditions, the particle morphology is mainly spherical and distributed evenly with an average diameter of 2 µm, and the gap between each particle is also approximately 2 µm. Based on the above assumptions for the critical diameter d_{pcrit}, we estimate that the particles can reach a maximum diameter of 4 µm if they can come into direct contact, which is consistent with the observed value of 2.6–6 µm by other researchers under chemical kinetics-controlled conditions [28]. Additionally, the pyrolysis coke sampled under our experimental conditions was found to have a C/H ratio varying from 2.49 to 5.81, which is much lower than that of soot particles in conventional flames with a C/H ratio of 8. Based on the measurement data, the value of the average surface site density can be calculated from formulas (4)–(7) and are shown in Table 2.

3.4. Simulation method and procedures

For a low Reynolds number (< 2300) under the experimental conditions, the turbulence in the tube was ignored. The numerical simulation was carried out on the Chemkin Pro platform, which

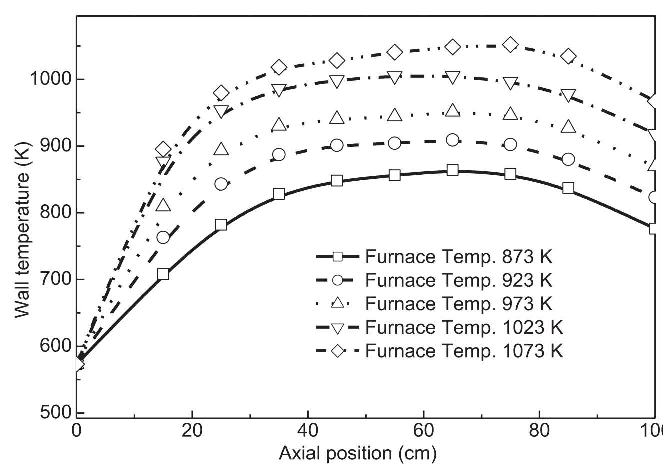


Fig. 7. Boundary conditions of the tube reactor at different furnace temperatures.

is chemical reaction analysis software, using a two-dimensional cylindrical shear flow (CSF) module. The temperature boundary conditions at the outer wall for every case were set as a function of the axial position, which were fitted from experimental data as shown in Fig. 7. The grid independence was checked, and the adopted grid number was 25 for the radial direction and 1000 for the axial direction.

In this work, the gaseous pyrolysis and surface reaction mechanisms were written in the Chemkin format and input into the software. The thermal and transport properties for every element were selected from previous studies [11,18,42]. The values of *SDEN* at different furnace temperatures were regarded as the input parameters and estimated in this work, as shown in Table 2. After the simulations, the distribution of species, the radicals, and the surface reaction rate in the tube, including the sensitivity analysis (SA) and ROP analysis for each reaction, can be obtained.

Table 2
Calculated results of *SDEN* in different coke growth stages.

Temp. (K)	ρ_p (g cm^{-3})	d_{pcrit} (μm)	C/H ratio	$SDEN_{w\text{-}s}$ (mol cm^{-2})	$SDEN_{w\text{-}sp}$ (mol cm^{-2})	$SDEN_{w\text{-}p}$ (mol cm^{-2})	$SDEN_{w\text{-}ave}$ (mol cm^{-2})
973	1.8	~4	2.49	3.05E−06	8.05E−06	3.89E−06	5.00E−06
993	1.8	~4	2.77	2.75E−06	7.26E−06	3.50E−06	4.50E−06
1013	1.8	~4	3.08	2.49E−06	6.55E−06	3.16E−06	4.07E−06
1023	1.8	~4	3.92	1.96E−06	5.17E−06	2.50E−06	3.21E−06
1033	1.8	~4	4.22	1.83E−06	4.81E−06	2.32E−06	2.99E−06
1053	1.8	~4	4.98	1.55E−06	4.09E−06	1.98E−06	2.54E−06
1073	1.8	~4	5.81	1.33E−06	3.51E−06	1.70E−06	2.18E−06

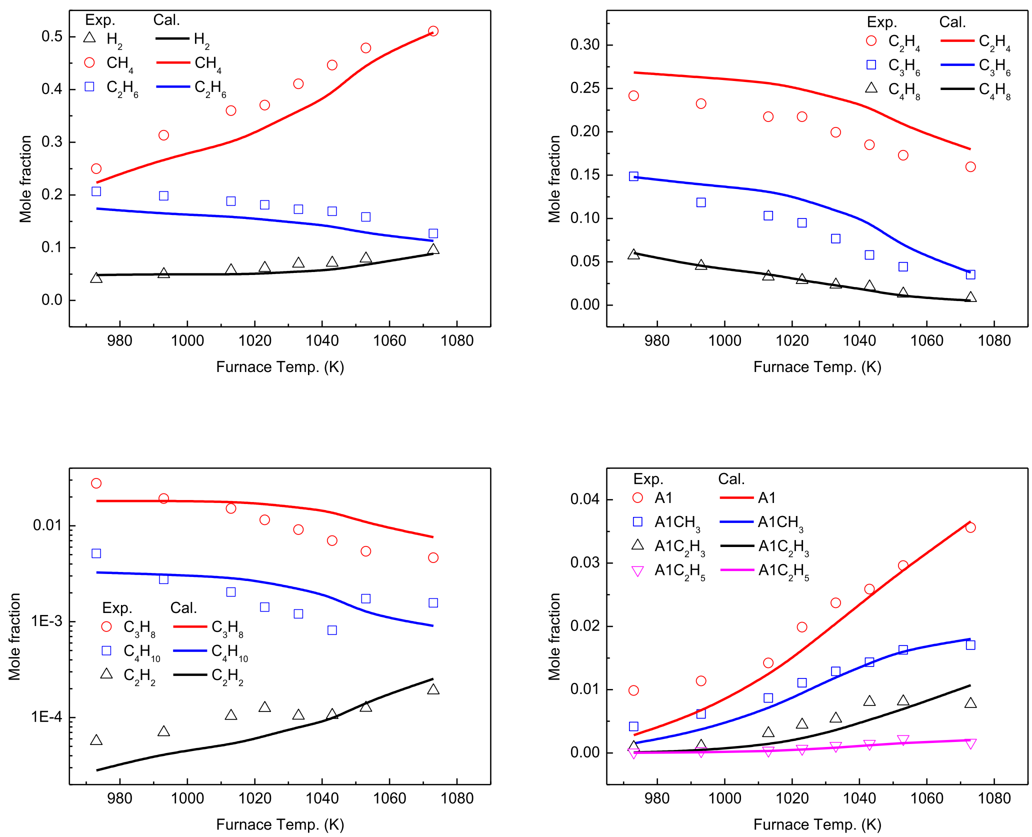


Fig. 8. Prediction results of pyrolysis products against experimental data.

4. Results and discussion

4.1. Model validations

4.1.1. Effect of temperature on the product concentrations

Under the conditions of an *n*-heptane inlet flow rate of 0.5 ml/min, an operating pressure of 1.0 MPa, and a furnace temperature of 973–1073 K, Fig. 8 shows the comparison of the experimental data and the simulation results at different furnace temperatures for some pyrolysis products. As shown in the figure, the mole fractions of C_2H_4, C_2H_6, CH_4, and C_3H_6 are the main gaseous products, and A1, $A1CH_3$, $A1C_2H_3$ and $A1C_2H_5$ are the main liquid products. With increasing temperature, the mole fractions of CH_4 and H_2 increase, while the C_2H_4, C_3H_6, and 1-C_4H_8 contents decrease sharply, and the concentrations of C_2H_6, C_3H_8, and C_4H_{10} also drop to some extent. However, for some unsaturated species with high C/H ratios, such as C_2H_2, A1, $A1CH_3$ and $A1C_2H_3$, their overall concentrations are relatively low but increase very rapidly as the temperature increases. A detailed discussion of the gaseous kinetic mechanism has been presented in previous works [16,39]. It can be seen from the figure that the gas phase reaction mechanism used in this work can well predict the variation in the key component concentration with temperature and can provide accurate reaction field data for the calculation and analysis of subsequent coking characteristics.

4.1.2. Effect of pyrolysis time on the coke formation rate

Under the conditions of an *n*-heptane inlet flow rate of 0.5 ml/min, an operating pressure of 1.0 MPa, and a furnace temperature of 1023 K, Fig. 9 shows the coke accumulation mass and its instantaneous formation rate with the change in the time duration. Based on the curve drawn from the experimental data, the coke formation rate first increases quickly to a high point (from A to B) and then drops to a lower value (from B to C). Consequently, the rate remains approximately stable for a certain duration (from C to D). In the end, the rate sharply increases until the pipe is blocked (from D to E). The change in the coke formation rate can be explained by our previous analysis of the active sites. Before the D point, the curve corresponds to the spatial growth, transition growth and plane growth stages. Using the estimation values of the active sites for each stage, $SDEN_{w\text{-}s}$, $SDEN_{w\text{-}sp}$ and $SDEN_{w\text{-}p}$ as shown in Table 2, we give the calculation results of the coking rate over time by the linear segmentation simulation method, which is also shown in Fig. 9. The calculation results can reasonably predict

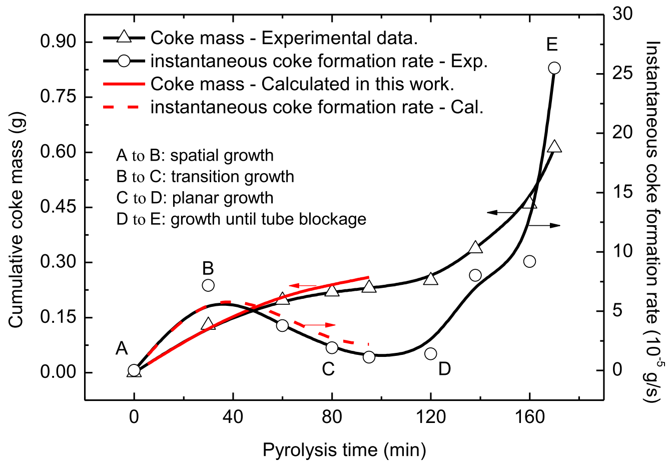


Fig. 9. Coke accumulation mass with the change in time duration (1.0 MPa, 1023 K).

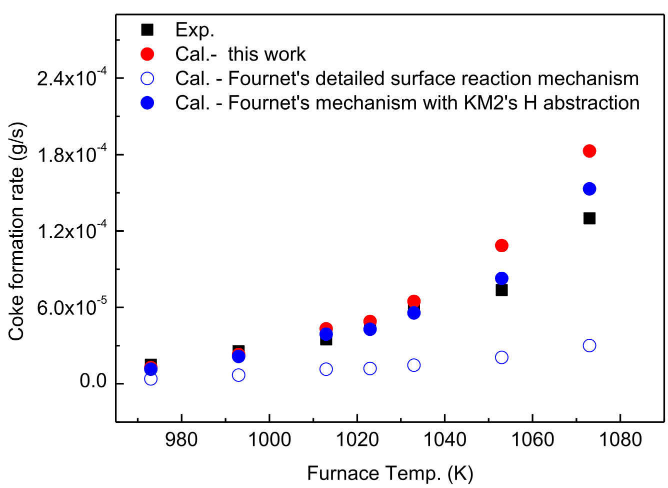


Fig. 10. Prediction of coke formation rate against experimental data.

the variation trend, and these results are consistent with the experimental data except for the last stage from D to E. At the growth stage near tube blockage, the explosive growth of the coking rate may be controlled by various factors, such as uneven temperature distribution in the pipeline, reduced flow area, increased pressure, prolonged gas phase residence time, and the contribution of soot formation, which require further study.

4.1.3. Effect of temperature on the coke formation rate

Under the conditions of an *n*-heptane inlet flow rate of 0.5 ml/min, an operating pressure of 1.0 MPa, a temperature from 973 to 1073 K, and a pyrolysis time of 80 min, Fig. 10 shows the prediction values of the average coke formation rate against the experimental data, and good agreement was obtained. The used value for surface site density is $SDEN_{w\text{-}ave}$, as shown in Table 2. With increasing temperature, the coke formation rate is enhanced. As mentioned in the above section, the main difference between this work and the detailed mechanism is the treatment for H abstraction reactions. We used the rate constants of the H, CH_3 and C_3H_3 abstraction reactions selected from the KM2 mechanism. As revealed by the figure, if the H abstraction reaction rate constants of the KM2 mechanism are substituted for the original detailed ones, the difference in the prediction rate from the simplified and detailed mechanism is very small, thus indicating two points: the coking rate is highly sensitive to the speed of hydrogen extraction by small molecule radicals, and the simplification of the detailed reaction mechanism is effective, at least for reliably predicting the coking rate. However, whether by a detailed mechanism or a simplified mechanism in this work, the prediction at high temperatures is higher than the experimental value, which may be related to the uncertainty of the reaction rate constants and the effect of the diffusion limitation, as discussed in the following sections.

4.2. Rate of production analysis for surface reactions in the flow reactor

4.2.1. Hydrogen abstraction reactions

The hydrogen abstraction reaction is the first and key step of the coking process, and it provides the effective addition position open(s) for unsaturated gaseous species; therefore, it is necessary to analyze the abstraction rate of various radicals. Under the chemical kinetics-controlled regime, the H abstraction reaction rate g_{rec} is expressed as follows:

$$g_{rec} = k_{rec} C_{R^*,g} Y_{H(s)} \cdot SDEN \cdot S \tag{8}$$

where $C_{R^*,g}$ is the concentration of a certain radical around the surface in the gas phase field (mol cm^{-3}), k_{rec} is the kinetic constant of the reaction (cm^3 mol^{-1} s^{-1}), $SDEN$ is the surface site density (mol cm^{-2}), $Y_{H(s)}$ is the proportion of the surface hydrogen and S is the contact area between the coke surface area and the gaseous radicals (cm^2). At surface temperatures of 973 and 1073 K, Fig. 11 shows the distribution of hydrogen abstraction rates of different radicals in the pipeline. It can be seen from the figure that the hydrogen abstraction rate of the H radical is the largest, followed by small molecule olefins and alkane radicals, such as CH_3, C_2H_3, C_2H_5, and C_3H_5, followed by the alkynyl groups, such as C_2H_2, C_3H_3, C_4H_3 and tolyl radicals, and finally, the abstraction rate of the larger radical C_9H_7 is very low. At low temperatures, as the main type of radicals in the gas phase, small olefins and alkyl radicals have a considerable ability to abstract hydrogen compared with H radicals. At high temperatures, the effects of unsaturated alkyne and aromatic hydrocarbon radicals are enhanced, while the action of olefins and alkane radicals is decreased, which is consistent with the change in the various groups of species at different temperatures, as shown in Fig. 8.

However, Fig. 11 shows the calculation results under the chemical kinetics-controlled regime, which implies the assumption that the mass diffusion rate of the gas phase component to the surface is infinite. In the actual growth process, the diffusion mechanism may have a certain impact. Considering the “spatial growth” process and the gaseous phase flow around a spherical particle, the mass balance equation between gaseous phase diffusion and reaction can be expressed as:

$$\begin{aligned} &K_{dif}\left(C_{R^*,g} - C_{R^*,s}\right) = K_{rec} C_{R^*,s} \\ &K_{dif} = D_{eff} Sh / d_c \\ &K_{rec} = k_{rec} \cdot Y_{H(s)} \cdot SDEN \\ &Sh = 2 + 0.552 Re^{1/2} Sc^{1/3} \end{aligned} \tag{9}$$

where K_{dif} is the mass transfer coefficient and K_{rec} is the chemical reaction coefficient (cm s^{-1}), D_{eff} is the effective diffusivity (cm^2 s^{-1}), Sh is the Sherwood number, d_c is the characteristic size (cm), which is equal to the particle diameter d_p, and $C_{R^*,s}$ is the concentration of radicals that actually reach the surface (mol cm^{-3}). The equations and methods for solving the diffusion parameters can be found in our previous work on the internal diffusion in a porous char [52]. Combining formulas (8) and (9), the actual abstraction rate with the diffusion effect can be obtained:

$$g_a = \left(\frac{K_{dif}}{K_{dif} + K_{rec}}\right) g_{rec} = \eta \cdot g_{rec} \tag{10}$$

The diffusion influence factor η can be used to characterize the extent of the diffusion effect. The closer the value is to 1, the

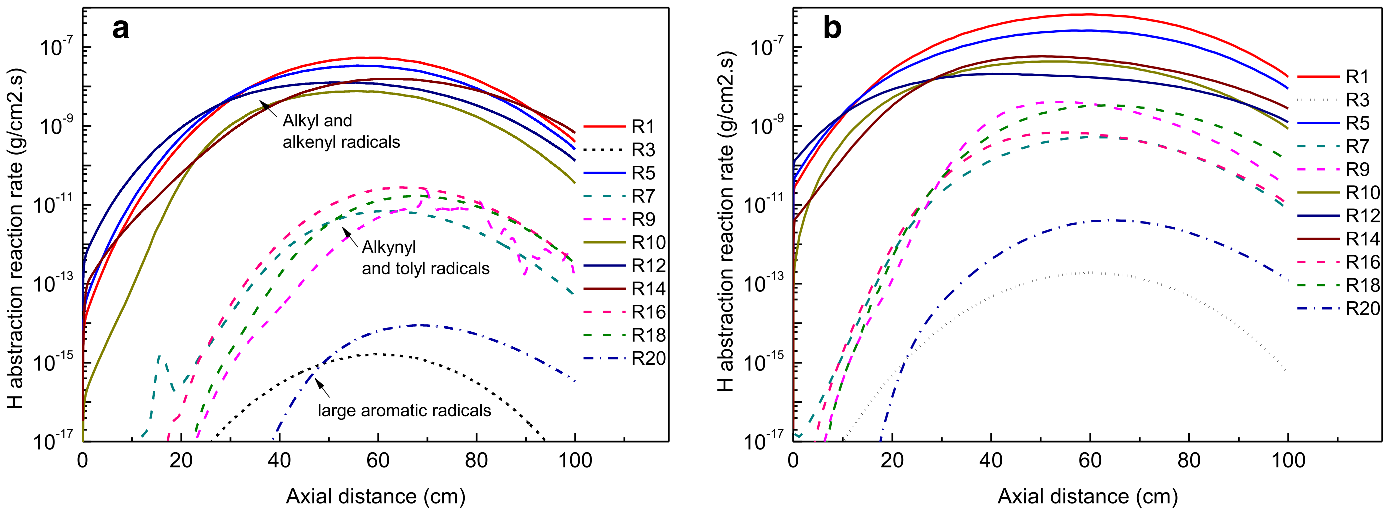


Fig. 11. Distribution of hydrogen abstraction rates of different radicals: (a) 973 K and (b) 1073 K.

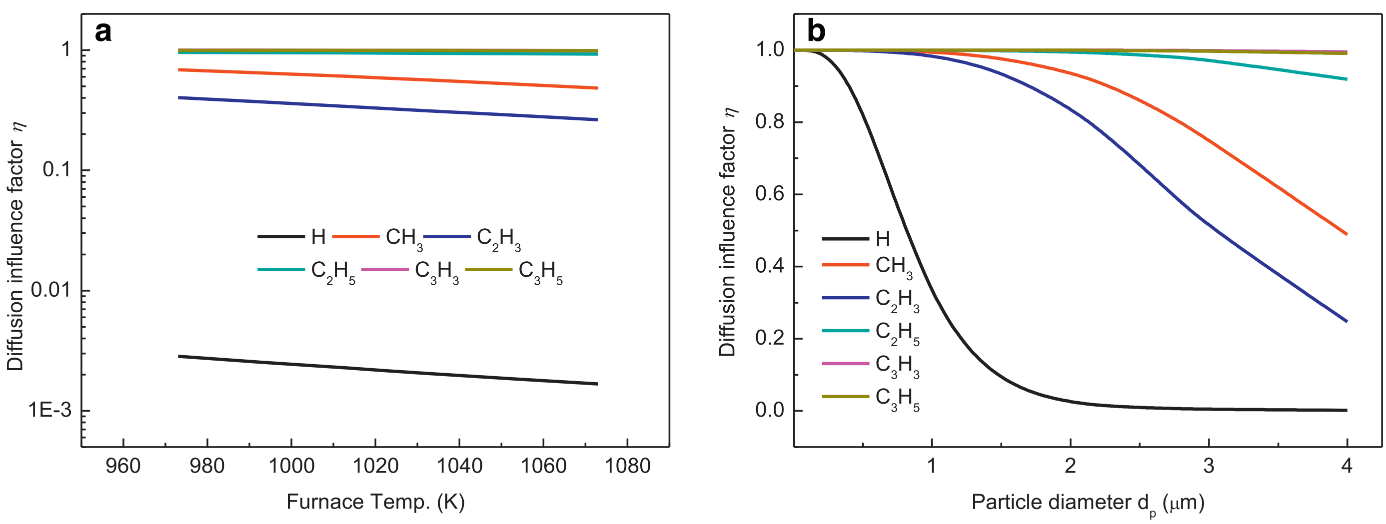


Fig. 12. Diffusion influence factors of H abstraction reactions at different temperatures and particle diameters: (a) at a particle diameter of 4 μm and (b) at a furnace temperature of 1023 K.

more likely the reaction is controlled by chemical kinetics. We selected several important radical reactions of H, CH_3, C_2H_3, C_2H_5, C_3H_3 and C_3H_5 and calculated the factors at different temperatures and particle diameters, as shown in Fig. 12. The effect of diffusion on the hydrogen abstraction reaction is enhanced with increasing temperature. Under the experimental temperature conditions selected in this work, the reaction rate of the H radical is strongly inhibited. Similarly, as a particle grows to the critical diameter defined in Section 3.3, the diffusion effect clearly reduces the abstraction ability of the H radical and then causes the coking speed to decelerate. However, due to the action of other radicals, such as CH_3, C_2H_3, and C_3H_5, the overall hydrogen extraction rate can be maintained at a relatively high level. At the same time, under the experimental conditions, the calculated results of the particles at 2–4 μm are consistent with the experimental observation in Fig. 6. After the diameter reaches 2 μm and the gap between the particles decreases, the diffusion limit becomes stronger. It is difficult for two single particles to come into direct contact by continuing to grow freely like a single particle. That is, the critical diameter $d_{pcrit} = 4\,\mu m$ assumed in the chemical kinetics-controlled state is difficult to achieve under practical conditions, causing the effective surface density at the "spatial growth" stage to be lower than the ideal value. Correspondingly, the predicted values in the next "transition growth" and "planar growth" stages also deviate a little from the expectations. The actual coking surface we observed at a pyrolysis time of 80 min is still rugged and grainy, while other researcher saw an almost smooth surface after a relatively short period of time (20–30 min) [28], which mainly because they used the experimental conditions with much larger fuel feeding rate and the flow in their tube reactor is fully-developed turbulent, leading to that the mass transfer coefficient K_{dif} is much larger and the coking growth is mainly controlled by the chemical kinetics. Adequate radical diffusion may make the particle fusion quickly and surface smoothing easier. The above analysis can explain why the calculation results are higher than the experimental values at high temperatures and at the normal growth stage.

4.2.2. Species addition reactions

Based on the chemical reaction mechanisms introduced in Section 3.2.2, the addition reaction rate of an unsaturated hydrocarbon can be calculated by:

$$g_{add} = k_{add} C_{M,g} Y_{open(s)} \cdot SDEN \cdot S \tag{11}$$

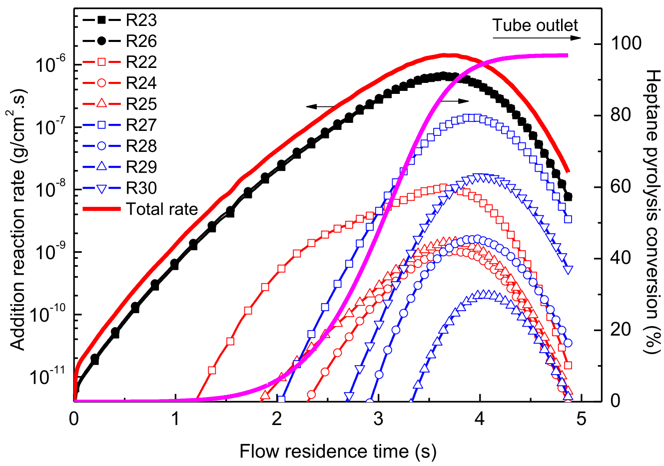


Fig. 13. Distribution of different addition reaction rates at a furnace temperature of 1023 K.

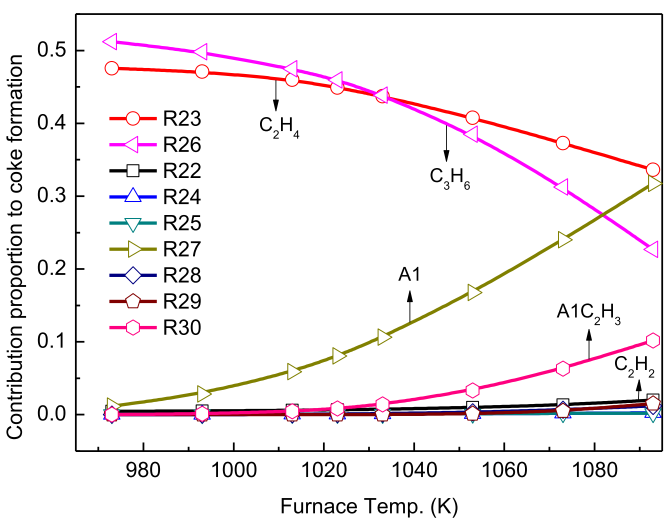


Fig. 14. Contribution of the species addition reaction to coke formation at different temperatures.

where $C_{M,g}$ is the concentration of a certain adding species around the surface in the gas phase field (mol cm^{-3}), k_{add} is the kinetic constant of the addition reaction (cm^3 mol^{-1} s^{-1}), and $Y_{open(s)}$ is the proportion of unoccupied active sites on the surface. Using a similar method to that introduced in the above section, the slight effect of the diffusion limit on the addition reactions was found because the value of $Y_{open(s)}$ is very low (down to 10^{-5}–10^{-6} magnitude), meaning that the whole addition reaction is chemical kinetics-controlled under the current experimental conditions.

Figure 13 shows the rate distribution of nine addition reactions at 1023 K. The position of the various species addition reactions varies greatly with the gas phase reaction time and heptane pyrolysis conversion. Consistent with their spatial concentration distributions, as the pyrolysis reaction proceeds, C_2H_4 and C_3H_6 first undergo a significant addition reaction, followed by small acetylenes and diolefins, including C_2H_2, PC_3H_4 and AC_3H_4. When the reaction time is long and at the outlet of the reactor, large amounts of aromatic species, such as A1, A2, $A1C_2H_3$, $A1C_2H$, are formed.

Figure 14 shows the contribution proportion to the total coke formation of these species under different temperatures. At lower temperatures, the addition of C_2H_4 and C_3H_6 contributes far more to the coke formation than do the other components. As the temperature increases, the contributions of C_2H_4 and C_3H_6 to the overall coking rate gradually decrease, while the contributions of small molecule aromatic hydrocarbons gradually increase. In particular, the contribution from A1 even exceeds the contribution of C_3H_6 at high temperatures above 1100 K. Smaller alkynes such as C_2H_2 have negligible addition contributions due to the small concentration in the gaseous phase, being only one thousandth of C_2H_4, under the experimental pyrolysis conditions.

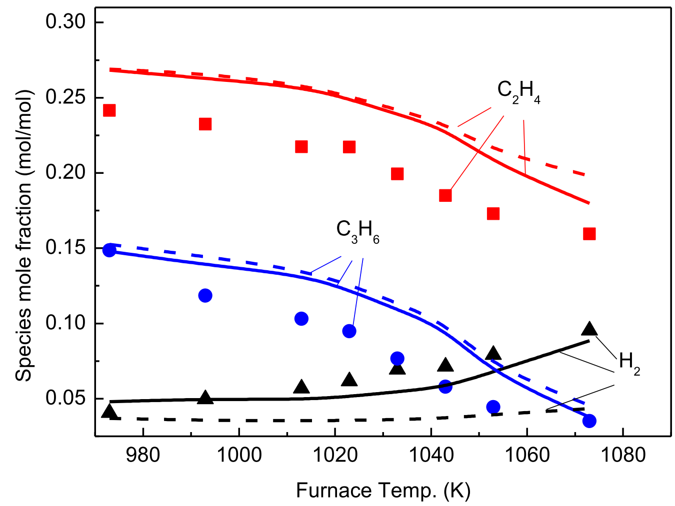


Fig. 15. Prediction of gaseous species concentration under the coke experimental conditions. (Solid symbols are experimental values, and solid and dashed lines represent the simulated results from the model with and without surface reactions, respectively).

The addition effects of unsaturated species increase the yield of the solid coke, and at the same time, a significant decrease in the species concentration in the gaseous phase occurs with increasing temperature. As shown in Fig. 15, the molar concentrations of C_2H_4 and C_3H_6 were overestimated by the model without considering the effect of surface reactions, and the simulated results considering the surface effect are closer to the experimental values. The solid particle is similar to an active carrier and promotes the separation of hydrogen and carbon in unsaturated hydrocarbons through the adsorption and reaction of small molecule radicals at their active sites, which ultimately leads to the formation of the more stable products, H_2 and coke. Both the experiments and the calculation with surface reactions prove the significant increase in the H_2 concentration in the gaseous phase.

5. Conclusion and remarks

To study the pyrolysis coking characteristics and mechanism of *n*-heptane in a millimeter-sized tube reactor, based on preliminary pyrolysis mechanism research, the *n*-heptane pyrolysis mechanism, PAH formation mechanism and wall surface reaction mechanism were developed, optimized and simplified. At the same time, based on the physical characteristics of coke particles, an estimation model of the surface site density was developed based on experimental phenomena. Combining the calculations in the Chemkin Pro software and the result of the coking experiments, the coking characteristics in the tube reactor were analyzed and discussed. The conclusions were drawn as follows:

(1) The coking precursor and gaseous radicals arise from the gaseous phase reaction, so the choice of the gas phase reaction mechanism and its prediction accuracy are important for studying coking characteristics. Based on the previous quantum calculation and experimental work, this study adjusted the first C–C bond dissociation reaction of *n*-heptane, added some reaction paths of benzene and toluene, and obtained a basic mechanism with 230 elements and 1127

chemical reactions for gaseous pyrolysis. The mechanism was verified by a comparison with the experimental data of the main pyrolysis products. Based on the detailed surface reaction mechanism for the CVD process, a simplification was carried out according to the theoretical analysis, and a simplified surface reaction mechanism with 34 steps suitable for this work was obtained. A comparison with the results of the coking experiment showed that the simplified and detailed mechanisms have little differences in predicting the coking rate, and the prediction accuracy can still be guaranteed with a greatly reduced computation time.

(2) Based on previous research on the characteristics of larger soot particles, we introduced the assumption of a critical diameter by observations and measurements of coke parameters, such as the coke particle diameter and C/H ratio, under the experimental conditions. For the three stages of particle spatial growth, particle fusion and planar growth, the formula for calculating the average surface site density was derived. The calculation results in the Chemkin Pro software agree well with the change in the coking rate with temperature and time under the experimental conditions.

(3) The hydrogen abstraction reaction is the first and key step of the wall surface reaction, which is generally considered to be driven by the active radicals in the gaseous phase [19,20]. In this work, the hydrogen abstraction reaction of ten radicals under pyrolysis conditions was calculated and compared. The results showed that the hydrogen abstraction reaction is affected by gaseous diffusion. As the temperature and particle diameter increase, the inhibition of diffusion becomes stronger. Among all the considered radicals, the abstraction ability of the H radical is the strongest, but it is easily restricted by diffusion. The presence of abundant radicals such as CH_3, C_2H_3, and C_3H_5 at low temperatures can compensate for this limitation, and they enable the surface hydrogen abstraction and subsequent addition reactions to maintain a certain rate.

(4) In this work, the addition reaction rates of nine unsaturated substances were analyzed. The calculation results showed that the addition of olefins, alkynes, diolefins and aromatic hydrocarbons occurs in sequence with increasing residence time. With increasing temperature, the addition contributions of olefins such as C_2H_4 and C_3H_6 gradually weaken, and the addition contributions of small molecule aromatic hydrocarbons such as A1 and $A1C_2H_3$ remarkably increase. Alkynes and diolefins, such as C_2H_2 and C_3H_4, have low concentrations under the experimental conditions, and their addition contributions are very low and can be ignored. As the degree of the coking reaction intensifies, the influence on the gas phase composition increases. Due to the addition of unsaturated substances, the concentration of H_2 in the gas phase is significantly improved.

(5) The calculation results and analytical conclusions in this work are based on the fact that the flow in the tube is not significantly affected by the coking body. Under actual conditions, prolonged pyrolysis can cause some abnormal flow conditions to occur. For example, a distinct throat will form at a certain position in the tube due to the non-uniform temperature and component distribution, and the vapor phase boundary layer may be destroyed due to continuous particle growth. The turbulent diffusion effect is remarkably enhanced when the tube diameter decreases sharply, and the pressure and residence time increase unexpectedly if the porous media flow is formed, resulting in pipe blockage. The study of these complicated processes requires the use of non-steady-state differential equations combined with the reaction mechanism for analysis and discussion.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2018.12.006.

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